

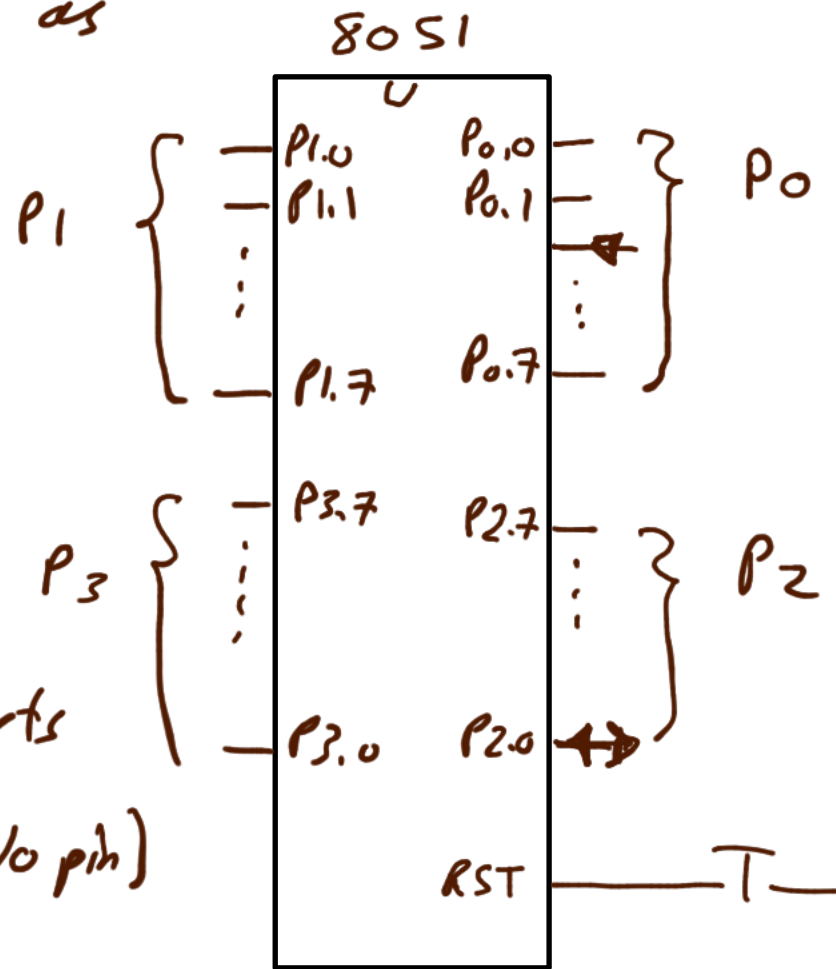
L # 8

I/O ports

- 8051 has 40 pin
- 32 pins are used as I/O pin
- each port is an 8 pin

$$\frac{32}{8} = 4 \text{ ports}$$

8051 has 4 I/O ports
(or $4 \times 8 = 32$ I/O pin)



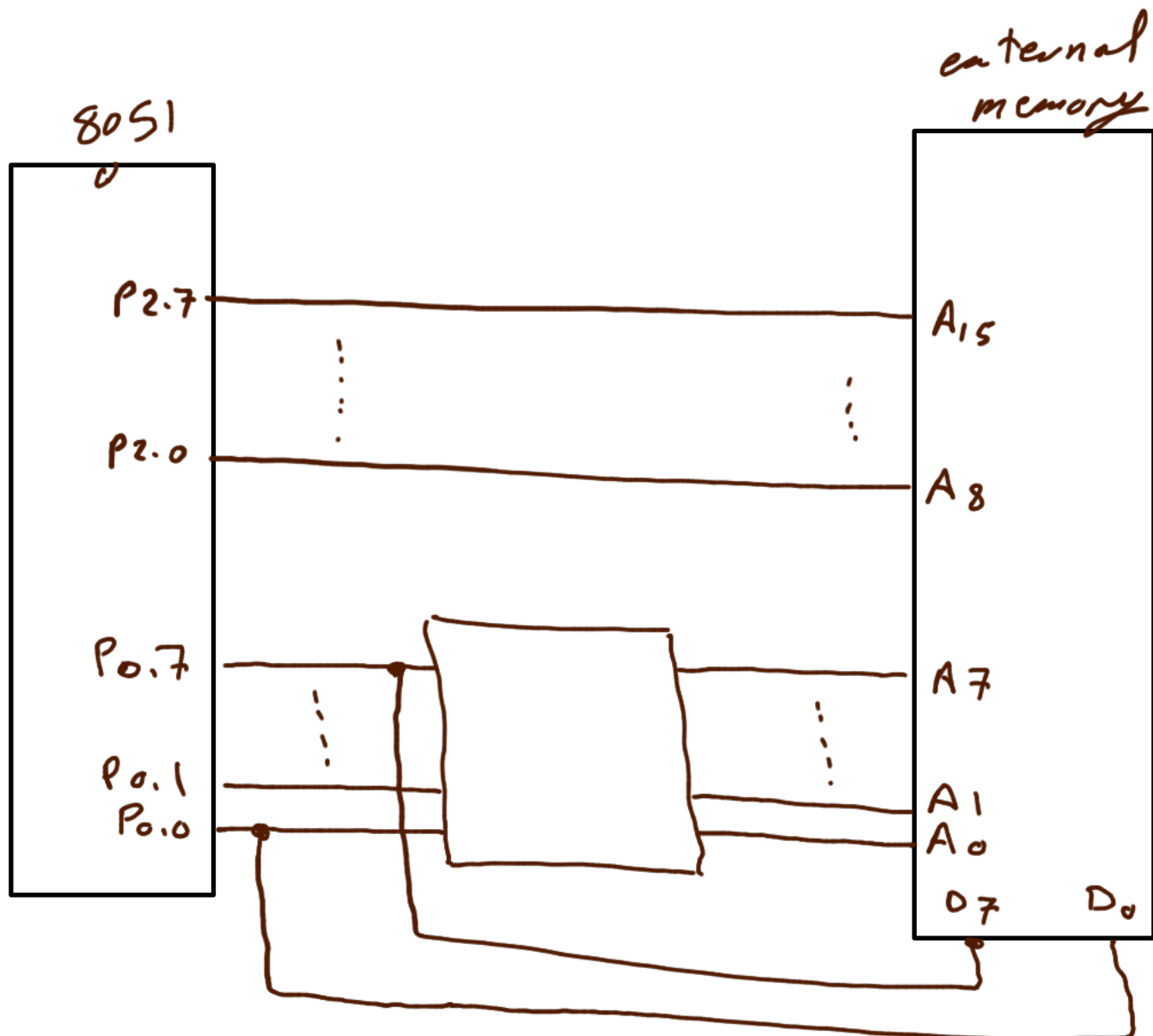
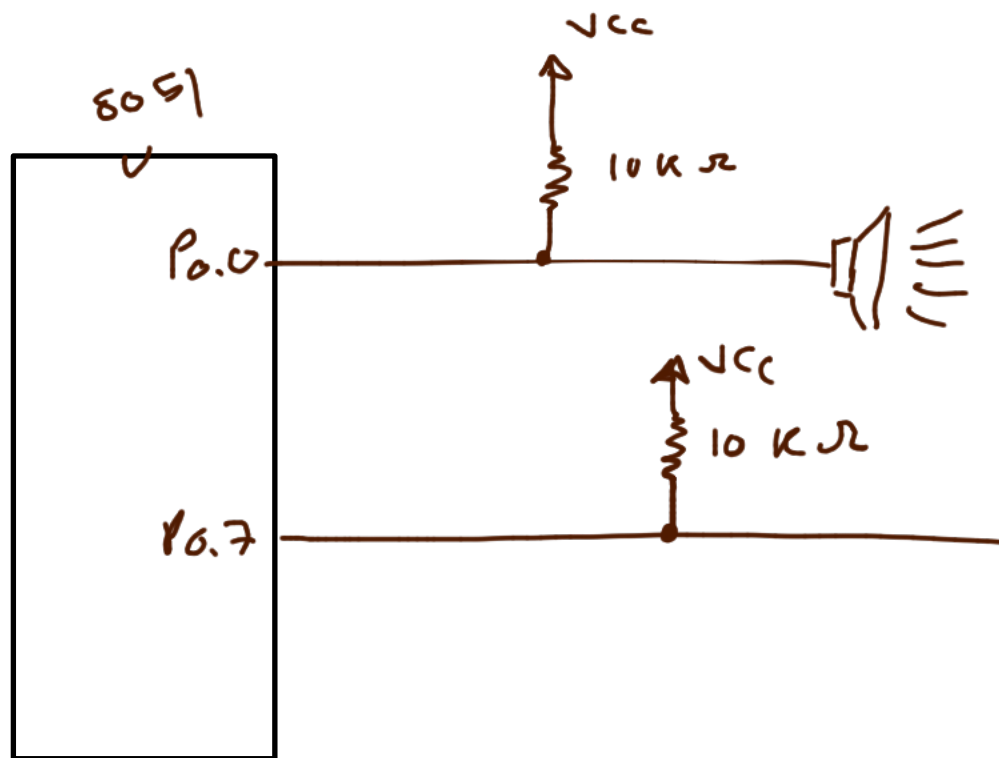
All ports (I/O pins) can be used as input/output

* port 0 $\Rightarrow$ P0

✓ It must be connected to pull up resistor $10 \text{ k}\Omega$

- is used to connect 8051 with external memory

low part of addresses $\longleftrightarrow$ P0
(AD0 — AD7)



* Port 1 $\Rightarrow P_1$

is used only as input/output
(no additional task).

* Port 2 $\Rightarrow P_2$

- is used to connect external memory
high part of addresses ($A_8 - A_{15}$)

* Port 3 $\Rightarrow P_3$

- each pin has special task

(slide #7)

$P_{3.6} \rightarrow \overline{WR}$ write signal } external memory
 $P_{3.7} \rightarrow \overline{RD}$ read "

slide #6

EA , $PSEN$, ALE

RST : Reset pin

Is used to terminate all activities and start with initial values (upon reset)

Register	value
A, B, R ₀ → R ₇	00H
PC, DPTR	0000H
SP	07H
P ₀ , P ₁ , P ₂ , P ₃	FFH

- By default all ports are input ports

- programmer can make any I/O pin as input or output.

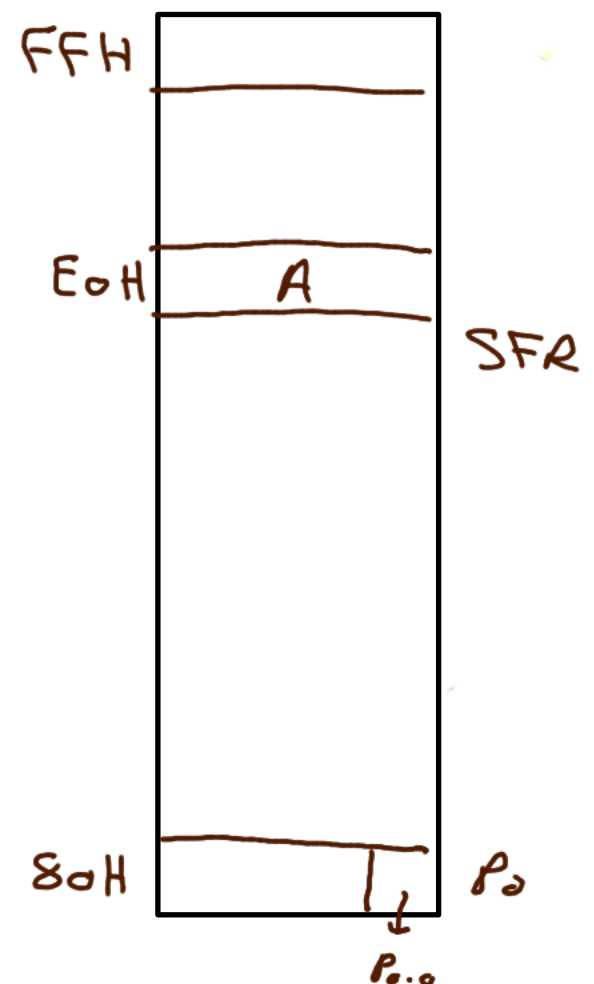
- To make any pin as input send 1 to it.

- " " " " " output " 0 to it.

Configure P3 as input

✓ MOV P3, # 1111 1111 B
 ↓ ↓
 P3.7 P3.0

✓ MOV P3, # 0 FFH

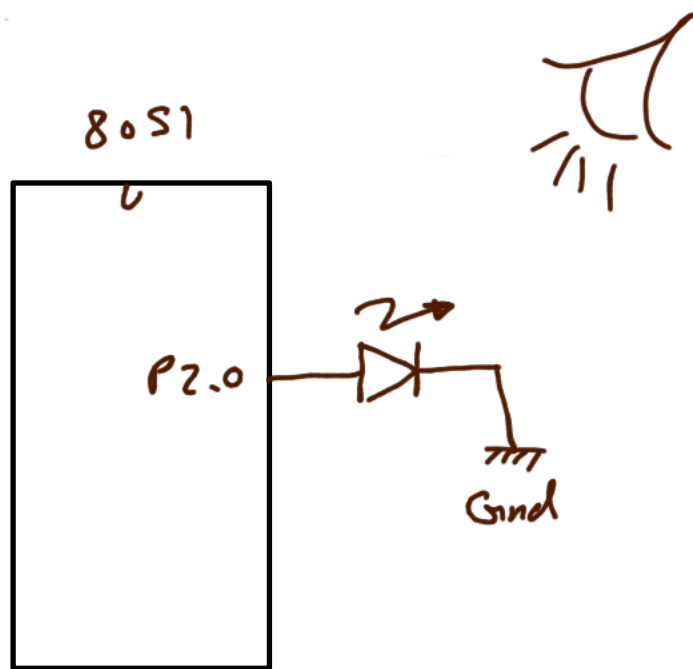


SETB P1.0 $\longrightarrow$ input pin

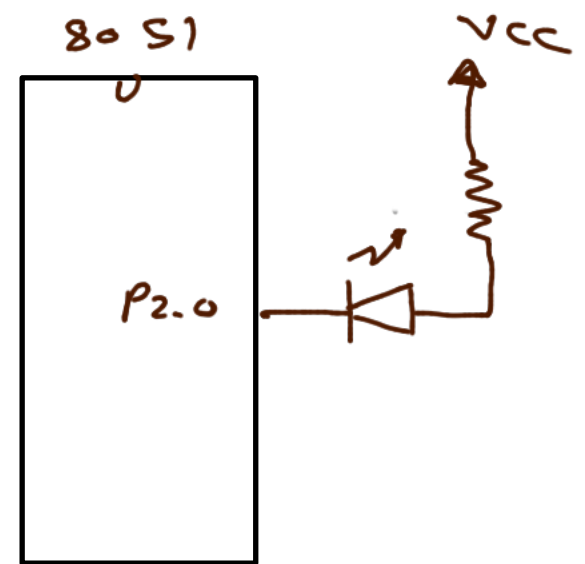
CLR P1.2 $\longrightarrow$ output pin

configure P2 to be output port

MOV P2, #00H ✓



SETB P2.0 ; ON LED
CLR P2.0 ; OFF LED



CLR P2.0 ; ON
SETB P2.0 ; OFF

6

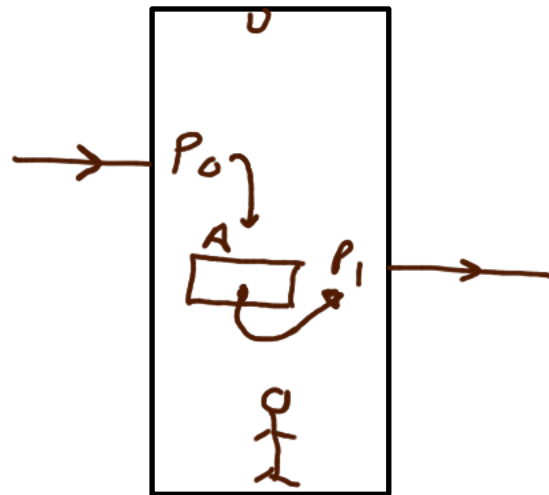
MOV P0, #0FFH ; P0 as input

CPL A

INC A

MOV PI, A

SJMP Back



complements

CPL A

MOV PI, A

MOV A, P2

CPL A

MOV P2, A

ACAU DE LAY

SSMP Back

DELAY: MOV R2, #200

HERE: DJNZ R2, HERE
RET

Do not use CPL

Back : MOV A, #55H

MOV PL, A

MOV P2, A

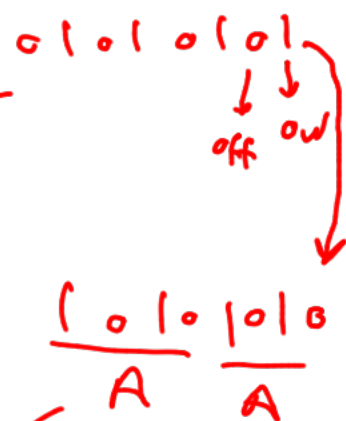
ACALL DELAY

MOV A, #0AAH

Mov R1, A

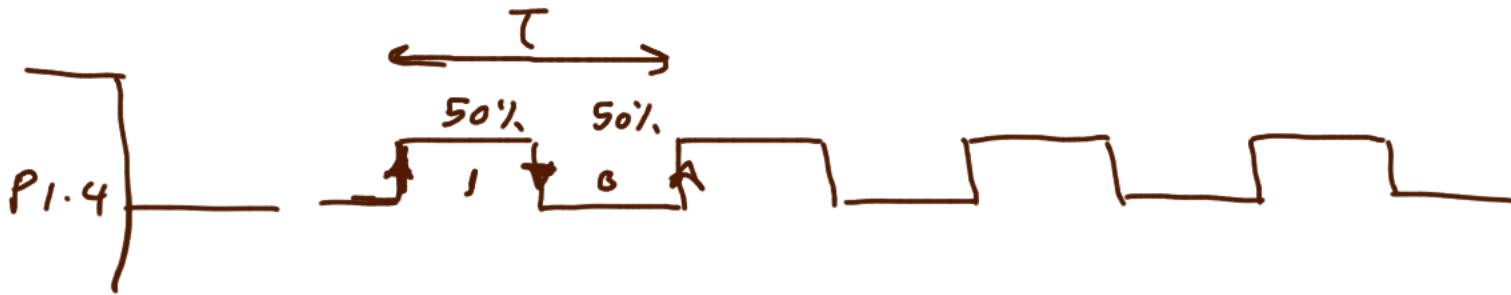
Mov P2, A

Stmp Back



write a program to create square wave of 50%
duty cycle on P1.4.

(7)



$$\text{Duty cycle} = \frac{\text{Time 1}}{T}$$

ORG 0

Back: SETB P1.4

ACALL DELAY ←

CLR P1.4

ACALL DELAY

SJMP Back

DELAY: MOV R4, #250

AGAIN: NOP

DJNZ R4, AGAIN

RET

Nop: no operation (do nothing)

update PC.

write a program to create wave of 75% duty (8)
cycle. on P2.7.



$$\frac{x}{y} = \frac{75}{25} = 3$$

$$x = 3y$$

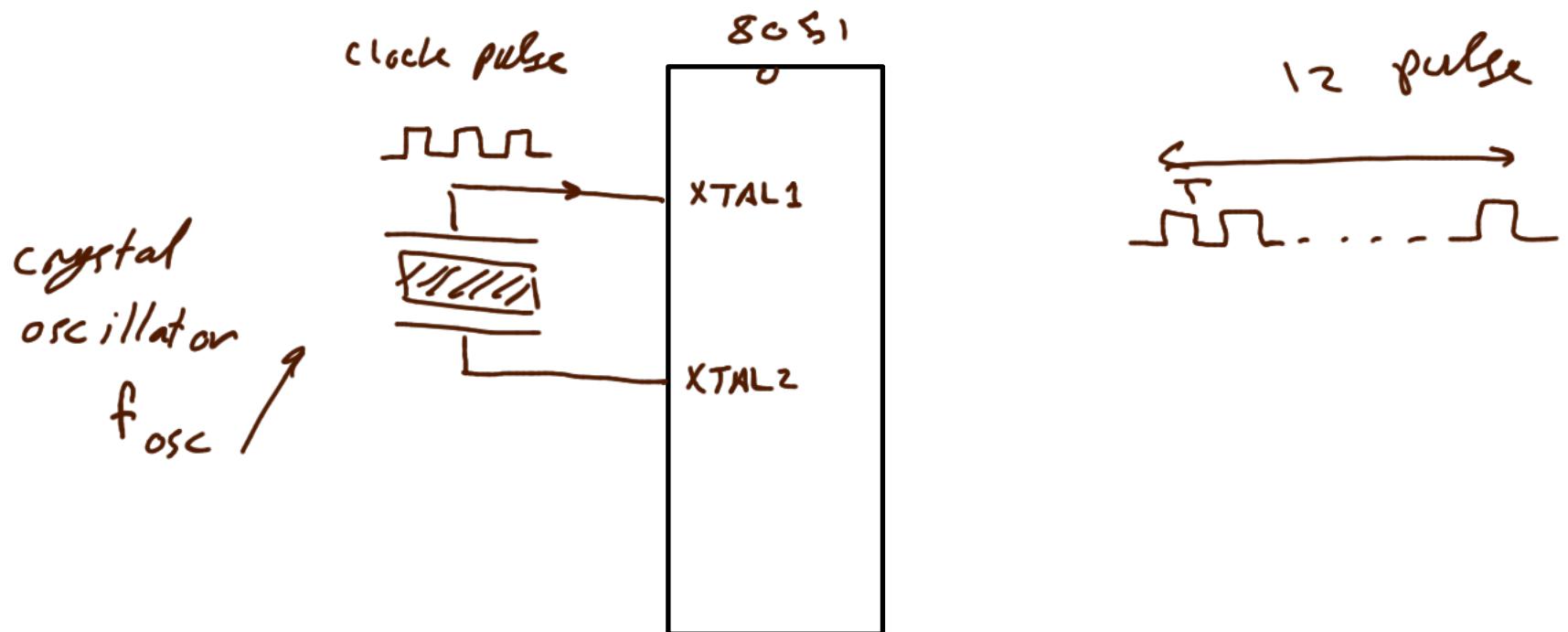
ORG 0

```
Back: SETB P2.7
      ACALL DELAY
      ACALL DELAY
      ACALL DELAY
      CLR P2.7
      -ACALL DELAY
      SJMP Back
```

66 %

$$\frac{x}{y} = \frac{66}{34} \approx 2$$

(9)



- 8051 divide crystal frequency by 12.

$$f_{sys} = \frac{f_{osc}}{12}$$

- $\text{MC} = \frac{12}{f_{osc}} \Rightarrow T_{sys} = MC = \frac{12}{f_{osc}}$

machine cycle

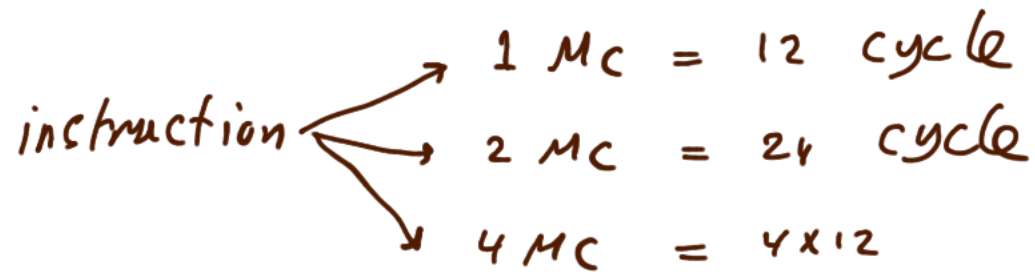
Assume $f_{osc} = 11.0592 \text{ MHz}$

$$f_{sys} = ?$$

$$MC = ?$$

$$f_{sys} = \frac{11.0592 \text{ MHz}}{12} = 0.9216 \text{ MHz}$$

$$MC = \frac{12}{11.0592 \text{ MHz}} = 1.085 \mu s$$



Assume crystal oscillator is 24 MHz, calculate the time of the following code.

	MC
MOV A, R4	1
NOP	1
NOP	1
MUL AB	4
MOV R2, B	1

$$MC = \frac{12}{f_{osc}} = \frac{12}{24 \text{ MHz}} = 0.5 \mu s$$

$$\begin{aligned} \text{Time} &= (1 + 1 + 1 + 4 + 1) \times MC \\ &= 8 \times 0.5 \mu s = 4 \mu s \end{aligned}$$

Assume XTAL = 6 MHz, compute time delay

DELAY: MOV R2, #100

NOP

HERE: DJNZ R2, HERE ↺

RET

MC	time
1	1 → 1 × 1 = 1
1	1 → 1 × 1 = 1
2	2 → 2 × 100 = 200
2	2 → 2 × 1 = 2

$$\begin{aligned}\text{Time delay} &= (1 + 1 + 200 + 2) \times MC \\ &= 204 \times 2 \mu s = \underline{408 \mu s}\end{aligned}$$

$$MC = \frac{12}{6 \text{ MHz}} = 2 \mu s \uparrow$$

HW #4

Q2 $f_{osc} = 12 \text{ MHz}$

$$MC = \frac{12}{f_{osc}} = \frac{12}{12 \text{ MHz}} = 1 \mu s$$

DELAY: CLR A

MOV R2, #4

MEZ: MOV R1, #50H

ME: MOV P1, A

CLR A

CLR A

NOP

DJNZ R1, ME

DJNZ R2, MEZ

RET

MC	Time
1	$1 \times 1 = 1$
1	$1 \times 1 = 1$
1	$1 \times 4 = 4$
1	$1 \times 320 = 320$
1	$1 \times 320 = 320$
1	$1 \times 320 = 320$
1	$1 \times 320 = 320$
2	$2 \times 320 = 640$
2	$2 \times 4 = 8$
2	$2 \times 1 = 2$

$$\text{Delay} = (1 + 1 + 4 + 320 + \dots + 2) \times MC$$

$$= (\quad) \times 1 \mu s$$

50H = 01010000
80

